

Supporting Information

Supporting Information for “Foundation or Formula? A Simulation-Based Comparison of Google’s TimesFM-3 and Classical Time-Series Models”

Ebrahim Khaled Ebrahim¹ | Somaia Mohamed Ali² | Ahmed El-Kotory³

¹Alexandria University, Egypt | ²Alexandria University, Egypt | ³Alexandria University, Egypt |

Correspondence: Ebrahim Khaled Ebrahim. ebrahimkhaled@alexu.edu.eg

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Summary

This file contains supporting information referred to in the main text as Sections S1 to S8.

S1 | Results under the mean absolute percentage error

Table S1 reports mean MAPE by process, alongside the percentage of series on which it cannot be computed. The D8 row illustrates the problem: every intermittent series contains a zero, so MAPE is undefined for all of them, and the process on which the error measures disagree most drops out of the comparison.

The rankings also move. On D4, MAPE places seasonal naive first (2.360) and TimesFM-3 second (2.801), whereas MASE places AutoARIMA first at every length from $n = 48$. MAPE's asymmetry between over- and under-forecasts is enough to reorder the methods on the same forecasts.

TABLE S1 | Mean MAPE over $h = 1, \dots, 12$ and percentage of series on which MAPE is undefined.

DGP	% undefined	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	TimesFM-3
D1	0%	2.785	2.737	2.681	2.336	2.501	2.283
D2	0%	3.291	3.341	3.277	2.696	2.971	2.641
D3	0%	9.509	9.278	9.875	10.078	9.393	9.214
D4	0%	2.360	7.471	4.986	2.862	4.538	2.801
D5	0%	2.971	3.583	3.270	2.436	2.916	2.256
D6	0%	2.011	2.379	1.835	1.902	1.925	1.822
D7	0%	7.634	4.631	2.979	2.538	2.767	2.184
D8	100%	—	—	—	—	—	—
D9	0%	1.861	1.769	1.741	1.470	1.599	1.457

Note: On the intermittent process D8 every series contains a zero, so MAPE cannot be computed for it. The best value in each row is in bold.

S2 | Monte Carlo standard errors

Table S2 gives the mean MASE of the six methods of the main design in every scenario, with its Monte Carlo standard error.

S3 | Coverage by process

Table S3 gives the empirical coverage of the nominal 80% intervals of the four foundation-model configurations and of AutoARIMA and AutoETS, by process.

S4 | Response-surface regressions

Tables S4 and S5 report the regressions summarised in the section of the main text on randomised parameters and departures from the assumptions.

S5 | Intermittent demand, parameter ranges and horizons

Table S6 gives the error measures of the six pre-registered methods on the intermittent process D8, and Table S7 compares TimesFM-3 with six Croston-family methods. Table S8 lists the parameter ranges of the randomised design, Figure S1 shows accuracy by forecast horizon, and Figure S2 shows how the relative accuracy of TimesFM-3 and AutoARIMA varies with the drawn parameters.

S6 | Supplementary diagnostics

Table S9 summarises each method's ratio to the best method of a scenario with and without the two scenarios in which only two seasonal cycles are observed (D4 and D5 at $n = 24$), and Table S10 gives the robustness summaries of the fifteen methods of Figure 5 of the main text, with bootstrap intervals. Table S11 lists the model forms chosen by AutoETS and AutoARIMA on the seasonal processes at $n = 24$ and $n = 48$, and Table S12 gives the 80% interval score by process.

TABLE S2 | Mean MASE and Monte Carlo standard errors, main design.

DGP	n	Seasonal naive	Theta	AutoETS	AutoARIMA	Combination	TimesFM-3
D1	24	1.646 (0.056)	1.682 (0.055)	1.568 (0.050)	1.485 (0.048)	1.538 (0.048)	1.417 (0.042)
D1	48	1.625 (0.058)	1.586 (0.055)	1.570 (0.054)	1.415 (0.043)	1.478 (0.048)	1.381 (0.039)
D1	96	1.626 (0.054)	1.572 (0.053)	1.564 (0.053)	1.324 (0.039)	1.435 (0.044)	1.307 (0.033)
D1	200	1.615 (0.061)	1.576 (0.057)	1.576 (0.057)	1.266 (0.035)	1.417 (0.046)	1.254 (0.033)
D2	24	1.905 (0.066)	2.003 (0.073)	1.886 (0.069)	1.708 (0.087)	1.788 (0.065)	1.675 (0.062)
D2	48	1.838 (0.068)	1.860 (0.069)	1.834 (0.068)	1.575 (0.053)	1.688 (0.058)	1.518 (0.042)
D2	96	1.810 (0.057)	1.812 (0.057)	1.811 (0.057)	1.517 (0.038)	1.643 (0.045)	1.455 (0.035)
D2	200	1.922 (0.067)	1.919 (0.067)	1.921 (0.067)	1.374 (0.034)	1.654 (0.050)	1.383 (0.033)
D3	24	4.411 (0.213)	4.483 (0.219)	4.892 (0.235)	5.275 (0.256)	4.659 (0.224)	4.656 (0.236)
D3	48	4.252 (0.208)	4.179 (0.206)	4.547 (0.207)	4.679 (0.239)	4.313 (0.203)	4.274 (0.205)
D3	96	4.125 (0.191)	4.135 (0.189)	3.938 (0.179)	3.940 (0.179)	3.871 (0.172)	4.023 (0.177)
D3	200	4.512 (0.210)	4.427 (0.211)	4.275 (0.199)	4.195 (0.201)	4.199 (0.199)	4.297 (0.198)
D4	24	1.029 (0.026)	4.146 (0.110)	3.602 (0.101)	2.284 (0.079)	3.084 (0.072)	1.321 (0.031)
D4	48	1.021 (0.024)	1.944 (0.048)	1.720 (0.032)	1.014 (0.030)	1.395 (0.028)	1.295 (0.027)
D4	96	1.005 (0.024)	2.910 (0.078)	2.407 (0.055)	0.886 (0.020)	1.839 (0.040)	1.124 (0.022)
D4	200	1.053 (0.022)	4.155 (0.117)	1.085 (0.024)	0.886 (0.021)	1.700 (0.043)	1.161 (0.020)
D5	24	1.139 (0.038)	3.154 (0.102)	3.025 (0.093)	1.574 (0.043)	2.474 (0.069)	1.152 (0.038)
D5	48	1.108 (0.032)	0.956 (0.038)	0.741 (0.027)	0.717 (0.024)	0.731 (0.023)	0.815 (0.029)
D5	96	1.076 (0.033)	0.757 (0.025)	0.761 (0.029)	0.732 (0.027)	0.713 (0.027)	0.761 (0.029)
D5	200	1.040 (0.030)	0.759 (0.020)	0.681 (0.024)	0.626 (0.022)	0.645 (0.022)	0.689 (0.024)
D6	24	0.801 (0.025)	1.327 (0.038)	0.766 (0.024)	0.802 (0.025)	0.837 (0.025)	0.807 (0.036)
D6	48	0.947 (0.028)	1.115 (0.031)	0.869 (0.023)	0.925 (0.027)	0.924 (0.025)	0.825 (0.022)
D6	96	1.006 (0.028)	0.963 (0.026)	0.905 (0.023)	0.920 (0.023)	0.916 (0.023)	0.865 (0.021)
D6	200	0.934 (0.023)	0.840 (0.017)	0.825 (0.017)	0.833 (0.017)	0.827 (0.017)	0.844 (0.019)
D7	24	5.417 (0.080)	3.179 (0.058)	1.272 (0.042)	1.083 (0.037)	1.680 (0.048)	1.142 (0.033)
D7	48	4.243 (0.067)	2.262 (0.048)	1.602 (0.061)	1.076 (0.051)	0.970 (0.038)	0.751 (0.022)
D7	96	1.655 (0.042)	0.821 (0.020)	1.305 (0.047)	1.329 (0.048)	1.097 (0.036)	0.959 (0.032)
D7	200	0.917 (0.026)	1.301 (0.031)	0.966 (0.028)	0.994 (0.030)	0.954 (0.027)	0.933 (0.025)
D8	24	1.019 (0.054)	0.999 (0.034)	1.047 (0.058)	0.996 (0.032)	1.010 (0.039)	0.780 (0.031)
D8	48	1.066 (0.066)	0.984 (0.036)	0.988 (0.033)	0.962 (0.024)	0.974 (0.028)	0.708 (0.030)
D8	96	0.932 (0.047)	0.923 (0.017)	0.926 (0.015)	0.924 (0.015)	0.923 (0.015)	0.683 (0.022)
D8	200	1.005 (0.052)	0.943 (0.017)	0.939 (0.015)	0.939 (0.015)	0.940 (0.015)	0.694 (0.023)
D9	24	1.547 (0.061)	1.518 (0.057)	1.454 (0.057)	1.372 (0.057)	1.404 (0.054)	1.321 (0.049)
D9	48	1.588 (0.056)	1.486 (0.054)	1.476 (0.053)	1.244 (0.041)	1.353 (0.046)	1.239 (0.039)
D9	96	1.430 (0.051)	1.342 (0.048)	1.333 (0.048)	1.098 (0.034)	1.213 (0.040)	1.104 (0.034)
D9	200	1.577 (0.056)	1.503 (0.053)	1.504 (0.053)	1.190 (0.032)	1.341 (0.042)	1.200 (0.035)

Note: Mean MASE over $h = 1, \dots, 12$ with its Monte Carlo standard error in parentheses (standard deviation over the 200 replications divided by $\sqrt{200}$).

TABLE S3 | Coverage of 80% prediction intervals by process.

Process	TimesFM-3	TimesFM-3 (eval.)	TimesFM-2.5	Chronos-Bolt	AutoARIMA	AutoETS
D1	0.817	0.819	0.787	0.814	0.777	0.852
D2	0.817	0.818	0.782	0.824	0.766	0.874
D3	0.704	0.707	0.703	0.721	0.706	0.618
D4	0.846	0.856	0.810	0.769	0.756	0.661
D5	0.818	0.830	0.799	0.833	0.724	0.686
D6	0.859	0.873	0.913	0.921	0.986	0.987
D7	0.845	0.848	0.894	0.968	0.833	0.765
D8	0.818	0.869	0.835	0.846	0.836	0.843
D9	0.807	0.809	0.783	0.810	0.760	0.825

Note: Empirical coverage of the nominal 80% interval $[q_{0.1}, q_{0.9}]$, averaged over lengths and replications; series-cluster bootstrap Monte Carlo SE at most 0.017.

Pooling all 180 tests of the $h = 1-12$ slice into one Benjamini–Hochberg family gives 134 significant comparisons, 115 won by TimesFM-3 (13–9 against AutoARIMA); pooling all 540 tests gives 132 and 113 (12–9), against 132 and 113 (11–9) with the pre-registered families.

S7 | A longer horizon

The main design forecasts 12 steps. To see whether its conclusions depend on that choice, the nine processes were regenerated with the same seeds at length $n + 48$ and forecast 48 steps ahead by the six pre-registered methods, TimesFM-2.5

TABLE S4 | Response-surface regressions of the robustness study, processes D1–D4.

Process	Term	Coefficient	95% interval	<i>p</i>	<i>R</i> ²
D1	version: est period	0.000	[-0.002, 0.002]	0.945	0.005
D1	version: heavy outliers	-0.032	[-0.061, -0.003]	0.029	0.005
D1	version: heavy tail	-0.022	[-0.048, 0.004]	0.091	0.005
D1	version: outliers	-0.010	[-0.031, 0.010]	0.334	0.005
D1	$\log_2 n$	0.005	[-0.009, 0.020]	0.490	0.005
D1	parameter phi	-0.018	[-0.036, -0.000]	0.045	0.005
D1	parameter sigma	0.005	[-0.011, 0.020]	0.535	0.005
D2	version: est period	-0.000	[-0.001, 0.001]	0.795	0.013
D2	version: heavy outliers	-0.025	[-0.052, 0.002]	0.066	0.013
D2	version: heavy tail	-0.016	[-0.041, 0.009]	0.207	0.013
D2	version: outliers	0.004	[-0.014, 0.021]	0.661	0.013
D2	$\log_2 n$	0.001	[-0.015, 0.016]	0.946	0.013
D2	parameter phi	-0.035	[-0.055, -0.015]	0.001	0.013
D2	parameter sigma	0.011	[-0.007, 0.029]	0.224	0.013
D2	parameter theta	0.003	[-0.014, 0.020]	0.726	0.013
D3	version: est period	0.000	[-0.004, 0.005]	0.876	0.008
D3	version: heavy outliers	0.026	[-0.032, 0.084]	0.381	0.008
D3	version: heavy tail	0.020	[-0.036, 0.075]	0.487	0.008
D3	version: outliers	-0.004	[-0.039, 0.032]	0.843	0.008
D3	$\log_2 n$	0.017	[-0.016, 0.050]	0.306	0.008
D3	parameter phi	-0.055	[-0.094, -0.016]	0.006	0.008
D3	parameter sigma	0.002	[-0.035, 0.039]	0.917	0.008
D3	parameter theta	0.010	[-0.027, 0.047]	0.598	0.008
D3	parameter drift	-0.003	[-0.040, 0.034]	0.869	0.008
D4	version: est period	-0.004	[-0.009, 0.001]	0.141	0.233
D4	version: heavy outliers	-0.217	[-0.261, -0.172]	0.000	0.233
D4	version: heavy tail	0.061	[0.028, 0.095]	0.000	0.233
D4	version: outliers	-0.219	[-0.260, -0.178]	0.000	0.233
D4	$\log_2 n$	0.238	[0.210, 0.266]	0.000	0.233
D4	parameter phi	-0.051	[-0.084, -0.018]	0.002	0.233
D4	parameter sigma	0.056	[0.022, 0.090]	0.001	0.233
D4	parameter Phi	0.098	[0.064, 0.132]	0.000	0.233
D4	parameter amp	-0.039	[-0.071, -0.008]	0.013	0.233

Note: One ordinary least-squares regression per process of the per-series $\log_2$ ratio of the MASE of TimesFM-3 to that of AutoARIMA on $\log_2 n$ (not standardised), the version (reference: random parameters with Gaussian innovations) and the drawn parameters, each standardised to mean 0 and standard deviation 1. Standard errors are clustered by parameter draw, because the five versions of a draw share its parameters and normal draws; 4 000 series per process (3 985 for D8, where series with a zero MASE denominator are excluded). Positive coefficients move the ratio against TimesFM-3.

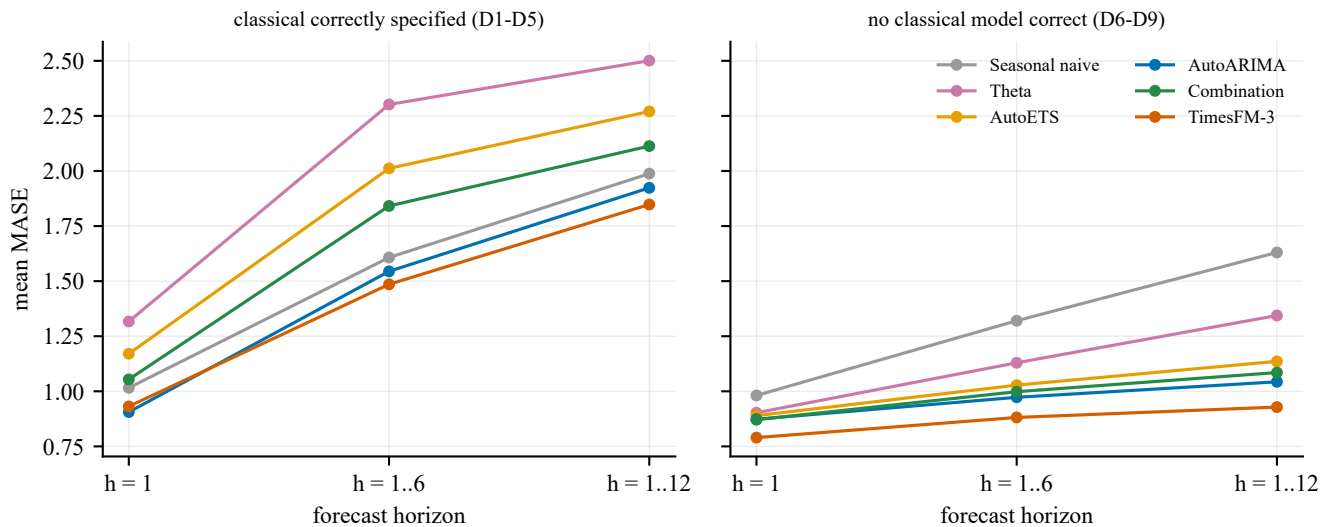


FIGURE S1 | Mean MASE by forecast horizon, averaged within each regime. Left: D1–D5, for which a classical family is correctly specified. Right: D6–D9, for which none is, or only in the conditional mean (D9).

TABLE S5 | Response-surface regressions of the robustness study, processes D5–D9.

Process	Term	Coefficient	95% interval	<i>p</i>	<i>R</i> ²
D5	version: est period	-0.086	[-0.111, -0.061]	0.000	0.073
D5	version: heavy outliers	-0.123	[-0.175, -0.071]	0.000	0.073
D5	version: heavy tail	-0.002	[-0.039, 0.036]	0.934	0.073
D5	version: outliers	-0.151	[-0.191, -0.111]	0.000	0.073
D5	$\log_2 n$	0.131	[0.104, 0.158]	0.000	0.073
D5	parameter sigma	0.039	[0.007, 0.072]	0.018	0.073
D5	parameter amp	-0.035	[-0.067, -0.003]	0.030	0.073
D5	parameter alpha	-0.001	[-0.034, 0.033]	0.969	0.073
D5	parameter beta	-0.014	[-0.048, 0.020]	0.422	0.073
D5	parameter gamma	0.028	[-0.005, 0.061]	0.100	0.073
D5	parameter trend	0.019	[-0.012, 0.051]	0.231	0.073
D6	version: est period	-0.002	[-0.004, 0.000]	0.066	0.020
D6	version: heavy outliers	-0.026	[-0.069, 0.018]	0.245	0.020
D6	version: heavy tail	0.018	[-0.022, 0.058]	0.384	0.020
D6	version: outliers	-0.023	[-0.053, 0.007]	0.140	0.020
D6	$\log_2 n$	-0.046	[-0.080, -0.013]	0.006	0.020
D6	parameter sigma	-0.031	[-0.071, 0.008]	0.123	0.020
D6	parameter jump sd	-0.031	[-0.065, 0.003]	0.071	0.020
D6	parameter break pos	-0.039	[-0.077, -0.001]	0.043	0.020
D6	parameter sigma level	0.033	[-0.002, 0.068]	0.064	0.020
D6	parameter jump sign	-0.019	[-0.054, 0.017]	0.295	0.020
D7	version: est period	-0.002	[-0.011, 0.008]	0.740	0.022
D7	version: heavy outliers	-0.074	[-0.136, -0.013]	0.017	0.022
D7	version: heavy tail	-0.031	[-0.081, 0.018]	0.216	0.022
D7	version: outliers	-0.045	[-0.094, 0.005]	0.075	0.022
D7	$\log_2 n$	0.031	[-0.010, 0.072]	0.138	0.022
D7	parameter sigma	-0.047	[-0.098, 0.005]	0.075	0.022
D7	parameter r	0.071	[0.020, 0.122]	0.006	0.022
D7	parameter infl pos	-0.086	[-0.140, -0.032]	0.002	0.022
D7	parameter K	0.010	[-0.040, 0.059]	0.703	0.022
D8	version: est period	-0.001	[-0.003, 0.001]	0.355	0.199
D8	version: heavy outliers	-0.093	[-0.125, -0.061]	0.000	0.199
D8	version: heavy tail	-0.047	[-0.078, -0.016]	0.003	0.199
D8	version: outliers	-0.025	[-0.043, -0.007]	0.007	0.199
D8	$\log_2 n$	-0.099	[-0.164, -0.034]	0.003	0.199
D8	parameter p	0.504	[0.417, 0.591]	0.000	0.199
D8	parameter lam	0.012	[-0.060, 0.084]	0.748	0.199
D9	version: est period	-0.002	[-0.003, -0.000]	0.045	0.001
D9	version: heavy outliers	-0.019	[-0.046, 0.008]	0.164	0.001
D9	version: heavy tail	-0.007	[-0.032, 0.018]	0.592	0.001
D9	version: outliers	-0.013	[-0.032, 0.007]	0.196	0.001
D9	$\log_2 n$	-0.002	[-0.017, 0.013]	0.807	0.001
D9	parameter phi	0.001	[-0.017, 0.020]	0.894	0.001
D9	parameter a	-0.000	[-0.019, 0.018]	0.974	0.001
D9	parameter b	-0.004	[-0.022, 0.014]	0.646	0.001

Note: One ordinary least-squares regression per process of the per-series $\log_2$ ratio of the MASE of TimesFM-3 to that of AutoARIMA on $\log_2 n$ (not standardised), the version (reference: random parameters with Gaussian innovations) and the drawn parameters, each standardised to mean 0 and standard deviation 1. Standard errors are clustered by parameter draw, because the five versions of a draw share its parameters and normal draws; 4 000 series per process (3 985 for D8, where series with a zero MASE denominator are excluded). Positive coefficients move the ratio against TimesFM-3.

and Chronos-2 (post hoc). Because the inflection of D7 sits at $0.6(n + H)$, the longer horizon moves it into the context for $n = 200$: the series has reached its plateau before the forecast origin. Table S14 and Figure S3 summarise the results.

S8 | Further checks

Table S15 compares count-data benchmarks with the history benchmark and TimesFM-3 on the intermittent process, Table S16 reports the tier of data observed after the documented training corpora, and Table S17 counts the scenarios separated by simultaneous and by per-scenario intervals.

TABLE S6 | Accuracy on the intermittent process D8 under four error measures.

Metric	n	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	TimesFM-3
MASE	24	1.019	0.999	1.047	0.996	1.010	0.780
	48	1.066	0.984	0.988	0.962	0.974	0.708
	96	0.932	0.923	0.926	0.924	0.923	0.683
	200	1.005	0.943	0.939	0.939	0.940	0.694
RMSSE	24	0.940	0.791	0.792	0.765	0.775	0.806
	48	0.937	0.746	0.726	0.718	0.724	0.774
	96	0.875	0.707	0.697	0.698	0.697	0.775
	200	0.923	0.718	0.706	0.706	0.706	0.797
sMAPE	24	86.696	172.741	169.988	165.908	170.489	191.774
	48	84.690	172.522	170.746	170.622	171.026	196.183
	96	84.171	170.801	169.574	170.065	169.949	197.610
	200	85.990	170.003	169.189	169.315	169.236	198.556
Pinball	24	0.634	0.395	0.402	0.380	0.388	0.325
	48	0.653	0.381	0.368	0.360	0.365	0.307
	96	0.614	0.348	0.341	0.341	0.341	0.288
	200	0.628	0.354	0.344	0.345	0.345	0.291

Note: Means over $h = 1, \dots, 12$ and 200 replications per length. The pinball loss is averaged over the nine deciles. Lower is better; the best value in each row is in bold.

TABLE S7 | Intermittent-demand baselines on the process D8 (post hoc analysis).

Metric	n	Croston	Croston-opt	Croston-SBA	ADIDA	IMAPA	TSB	AutoARIMA	TimesFM-3
MASE	24	1.042	1.024	1.025	0.972	0.972	0.971	0.996	0.780
	48	0.990	0.992	0.974	0.947	0.954	0.966	0.962	0.708
	96	0.935	0.948	0.921	0.926	0.923	0.916	0.924	0.683
	200	0.947	0.948	0.934	0.941	0.942	0.948	0.939	0.694
RMSSE	24	0.769	0.758	0.763	0.753	0.757	0.764	0.765	0.806
	48	0.722	0.726	0.718	0.717	0.722	0.733	0.718	0.774
	96	0.696	0.702	0.694	0.700	0.701	0.716	0.698	0.775
	200	0.709	0.710	0.707	0.709	0.712	0.730	0.706	0.797

Note: Mean MASE and RMSSE over $h = 1, \dots, 12$ and 200 replications per length. Croston-opt is Croston's method with optimised smoothing parameters and Croston-SBA the Syntetos-Boylan approximation. Lower is better; the best value in each row is in bold.

TABLE S8 | Parameter ranges of the randomised-parameter design.

ID	Parameters, each drawn uniformly on the stated interval
D1	$\phi \in [0.10, 0.95], \sigma \in [1, 5]$
D2	$\phi \in [0.10, 0.90], \theta \in [-0.6, 0.8], \sigma \in [1, 5]$
D3	$\phi \in [-0.5, 0.8], \theta \in [-0.5, 0.6], \text{drift} \in [0, 0.6], \sigma \in [1, 5]$
D4	$\phi \in [0.1, 0.8], \Phi \in [0.1, 0.8], \text{seasonal amplitude} \in [5, 20], \sigma \in [1, 5]$
D5	$\alpha \in [0.05, 0.5], \beta \in [0.001, 0.03], \gamma \in [0.05, 0.3], \text{initial trend} \in [0, 0.5], \text{amplitude} \in [5, 20], \sigma \in [1, 5]$
D6	$\text{jump} \in [3, 15] \sigma \text{ with random sign, break at } [0.50, 0.85] n, \text{level noise} \in [0.2, 1], \sigma \in [1, 5]$
D7	$r \in [0.04, 0.15], \text{inflection at } [0.4, 0.8](n + H), K \in [100, 250], \sigma \in [1, 6]$
D8	demand probability $p \in [0.1, 0.6]$, size $1 + \text{Pois}(\lambda)$ with $\lambda \in [1, 8]$
D9	$\phi \in [0.1, 0.9], a \in [0.05, 0.15], b \in [0.75, 0.90]$; if $a + b \geq 0.98$, b is set to $0.97 - a$

Note: Parameters not listed are as in Table 2 of the main text. The intervals contain the fixed values of the main design. The rule for D9 keeps the conditional variance stationary and puts a small point mass at $b = 0.97 - a$. Seeds are disjoint from those of the main design; the five versions share the parameter draws and, except for the heavy-tailed D8 sizes, the underlying normal draws.

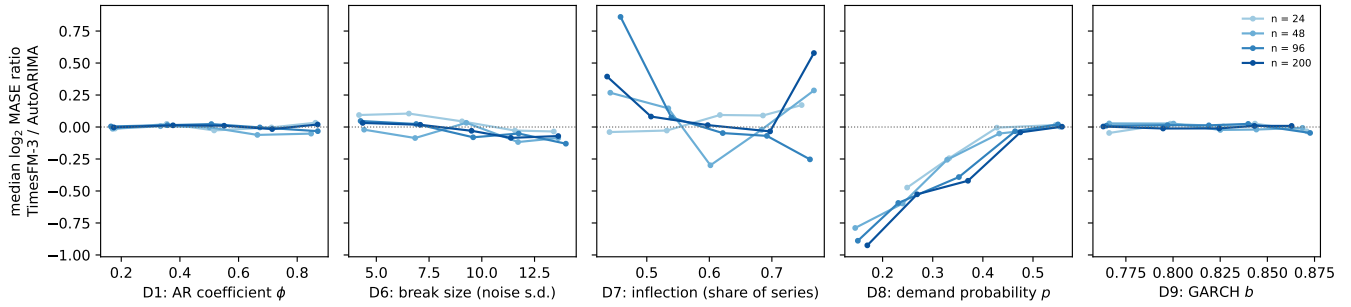


FIGURE S2 | Median $\log_2$ ratio of the MASE of TimesFM-3 to that of AutoARIMA, by quintile of the drawn parameter, randomised-parameter design with Gaussian innovations. Values below zero favour TimesFM-3. For D8 the ratio compares median-type forecasts and is shown for completeness only.

TABLE S9 | Ratio of each method's mean MASE to the best in the same scenario, with and without the two-cycle scenarios.

Method	Median	Geom. mean	90th pct.	Worst	Worst excl.	Scenario of worst excl.
Seasonal naive	1.200	1.331	1.60	5.65	5.65	D7, $n = 48$
Theta	1.251	1.458	2.97	4.69	4.69	D4, $n = 200$
AutoETS	1.194	1.296	1.91	3.50	2.72	D4, $n = 96$
AutoARIMA	1.032	1.118	1.37	2.22	1.62	D7, $n = 96$
Combination	1.108	1.228	1.74	3.00	2.08	D4, $n = 96$
TimesFM-3	1.008	1.050	1.22	1.31	1.31	D4, $n = 200$

Note: Main design, 36 scenarios, $h = 1, \dots, 12$. "Excl." leaves out D4 and D5 at $n = 24$, where only two seasonal cycles are observed; the 90th percentile is taken over the 36 scenarios.

TABLE S10 | Robustness of fifteen methods across the 36 scenarios of the main design.

Method	Worst ratio to best	Mean ratio to best	Within 10% (%)	Scenarios won
TimesFM-3, evaluator settings	1.27 [1.26, 1.33]	1.058 [1.055, 1.066]	81	5 [2, 8]
TimesFM-3	1.31 [1.29, 1.36]	1.063 [1.060, 1.072]	78	0 [0, 4]
TimesFM-2.5	2.18 [2.04, 2.33]	1.121 [1.116, 1.130]	67	7 [2, 8]
TiRex	2.24 [2.06, 2.45]	1.121 [1.116, 1.131]	64	0 [0, 4]
TimesFM-2.5, settings off	2.26 [2.10, 2.44]	1.124 [1.120, 1.134]	67	3 [0, 5]
AutoARIMA	2.22 [2.08, 2.39]	1.131 [1.125, 1.142]	64	8 [5, 10]
Chronos-2	2.75 [2.56, 2.97]	1.139 [1.135, 1.150]	67	4 [3, 6]
Combination (ETS, ARIMA, DOTM)	2.99 [2.84, 3.16]	1.241 [1.236, 1.253]	47	1 [0, 2]
Combination (Theta, ETS, ARIMA)	3.00 [2.85, 3.17]	1.243 [1.237, 1.255]	44	1 [1, 3]
Chronos-Bolt	3.06 [2.95, 3.25]	1.296 [1.291, 1.309]	58	1 [0, 2]
AutoETS	3.50 [3.28, 3.73]	1.312 [1.304, 1.327]	31	1 [0, 2]
Seasonal naive	5.70 [5.47, 6.03]	1.347 [1.337, 1.362]	25	2 [1, 3]
DOTM	4.66 [4.34, 4.99]	1.413 [1.405, 1.429]	22	1 [1, 2]
M4 Comb	4.45 [4.18, 4.79]	1.456 [1.447, 1.474]	19	1 [0, 2]
Theta	4.69 [4.37, 5.03]	1.476 [1.467, 1.493]	19	1 [1, 2]

Note: Main design, 7 200 series, mean MASE over $h = 1, \dots, 12$, ratios to the best of the fifteen methods in each of the 36 (process, length) scenarios. Brackets: 95% intervals from 500 bootstrap resamples of the replications within each scenario. Mean ratio is the geometric mean over scenarios. Intervals for the original six methods from 2000 paired resamples are given in Section 5.1 of the main text. Scenarios won: the two TimesFM-3 configurations split the scenarios they win (among the original six methods TimesFM-3 wins 16), and ratios can differ slightly from Table 3 of the main text because the best method of a scenario is taken over fifteen methods (seasonal naive: 5.70 against 5.65).

TABLE S11 | Model forms chosen by AutoETS and AutoARIMA on the seasonal processes, given $m = 12$.

Process	n	ETS seas. (%)	ARIMA seas. (%)	Most frequent ETS forms	Most frequent ARIMA orders
D4	24	8.5	0.0	ETS(A,N,N) (100), ETS(M,N,N) (83), ETS(A,N,A) (8)	(2,0,1)(0,0,0) (52), (3,0,1)(0,0,0) (44)
D4	48	98.0	100.0	ETS(A,N,A) (67), ETS(M,N,M) (60), ETS(M,N,A) (42)	(1,0,0)(1,1,0) (52), (1,0,0)(0,1,1) (26)
D5	24	0.0	0.0	ETS(A,N,N) (117), ETS(M,N,N) (83)	(2,0,1)(0,0,0) (72), (4,0,0)(0,0,0) (52)
D5	48	100.0	100.0	ETS(M,N,M) (54), ETS(A,A,A) (40), ETS(M,A,A) (26)	(0,1,1)(0,1,1) (48), (0,0,0)(0,1,1) (35)

Note: 200 replications per row, the series of the main design; counts out of 200 in parentheses. A form is seasonal when it has a seasonal component (ETS) or a non-zero seasonal order (ARIMA).

TABLE S12 | Scaled 80% interval score by process, averaged over the four lengths.

Process	Seasonal naive	Theta	AutoETS	AutoARIMA	Combination	TimesFM-3
D1	8.49	7.80	7.67	6.36	6.97	6.05
D2	9.05	9.28	9.09	6.98	8.00	6.86
D3	23.33	23.31	22.12	22.03	21.01	21.24
D4	4.58	16.77	11.04	5.65	9.65	5.69
D5	4.64	7.22	6.51	4.29	5.59	3.98
D6	9.35	7.13	7.21	7.53	7.28	4.55
D7	14.14	8.59	5.91	5.17	5.51	4.55
D8	10.11	4.60	4.50	4.40	4.47	3.54
D9	8.56	7.22	7.07	5.82	6.44	5.67

Note: Interval score of $[q_{0.1}, q_{0.9}]$ with $\alpha = 0.2$ (width plus $2/\alpha$ times the distance by which the outcome falls outside the interval), averaged over the horizon and scaled by the MASE denominator, as for MSIS; lower is better, the best in bold. TimesFM-3 has the lowest score in 22 of the 36 (process, length) scenarios.

TABLE S13 | Classical methods in forecast (R) and StatsForecast (Python) on the seasonal processes: mean MASE over $h = 1, \dots, 12$.

Process	n	R ARIMA	SF ARIMA	R ETS	SF ETS	R Theta	SF Theta	R seas. (%)
D4	24	2.866	2.284	3.949	3.602	4.140	4.146	0 / 13
D4	48	1.008	1.014	1.765	1.720	1.949	1.944	100 / 100
D4	96	0.891	0.886	1.044	2.407	2.896	2.910	100 / 100
D4	200	0.894	0.886	1.056	1.085	4.085	4.155	100 / 100
D5	24	1.695	1.574	3.304	3.025	3.147	3.154	0 / 0
D5	48	0.717	0.717	0.739	0.741	0.947	0.956	100 / 100
D5	96	0.747	0.732	0.719	0.761	0.759	0.757	100 / 100
D5	200	0.631	0.626	0.644	0.681	0.760	0.759	100 / 100

Note: The same 200 series per row. R: `auto.arima`, `ets` and `thetaf` of forecast 9.0.2 at their defaults, with frequency 12; SF: StatsForecast 2.1.1 as in the main text. R seas.: share of series given a seasonal form by `auto.arima` / `ets`. No R fit failed. On D4 at $n = 96$ StatsForecast's AutoETS settles on a near-zero seasonal smoothing parameter with a much lower likelihood than R's fit of the same form.

TABLE S14 | Longer horizon ($H = 48$): worst ratio to the scenario-best and 80% coverage by block of steps.

Method	Worst ratio, D7 excluded			Coverage 80%			D7, $n = 200$ MASE 1–48
	1–12	13–24	25–48	1–12	13–24	25–48	
TimesFM-3	1.31	1.39	1.80	0.788	0.774	0.760	8.37
TimesFM-2.5	1.42	1.59	1.77	0.781	0.734	0.707	12.22
Chronos-2	2.94	2.71	3.33	0.787	0.762	0.761	5.30
AutoARIMA	2.11	1.60	1.41	0.760	0.739	0.720	2.51
AutoETS	3.49	2.17	1.67	0.779	0.790	0.784	2.35
Theta	4.65	3.66	6.27	0.731	0.757	0.768	2.73
Combination	2.92	2.00	2.17	0.788	0.812	0.814	2.22
Seasonal naive	1.66	1.88	1.87	0.805	0.780	0.753	1.11

Note: The nine processes regenerated with the same seeds at length $n + 48$, 200 replications per scenario; ratios to the best of the eight methods in each of the 32 (process, length) scenarios without D7. Coverage is averaged over all 36 scenarios. The last column is the mean MASE over the 48 steps on D7 at $n = 200$, where the process has reached its plateau before the forecast origin.

TABLE S15 | Count-data benchmarks on the intermittent process D8.

Method	Scaled pinball loss				RMSSE of the mean			
	24	48	96	200	24	48	96	200
iETS	0.306	0.298	0.283	0.287	0.737	0.712	0.693	0.704
Negative binomial	0.320	0.304	0.289	0.295	0.744	0.710	0.692	0.703
TSB compound	0.330	0.314	0.298	0.306	0.764	0.732	0.714	0.729
History: deciles; mean	0.314	0.298	0.282	0.286	0.744	0.710	0.692	0.703
TimesFM-3 (mean of deciles)	0.325	0.307	0.288	0.291	0.760	0.717	0.694	0.705

Note: 200 replications per length (iETS failed on 3 series at $n = 24$). iETS: `smooth:adam(y, "MNN", occurrence = "auto")` with simulated quantiles; negative binomial fitted by maximum likelihood to the context; TSB compound: TSB occurrence probability ($\alpha = 0.2$) times the empirical distribution of the non-zero sizes. Point forecasts are the predictive means.

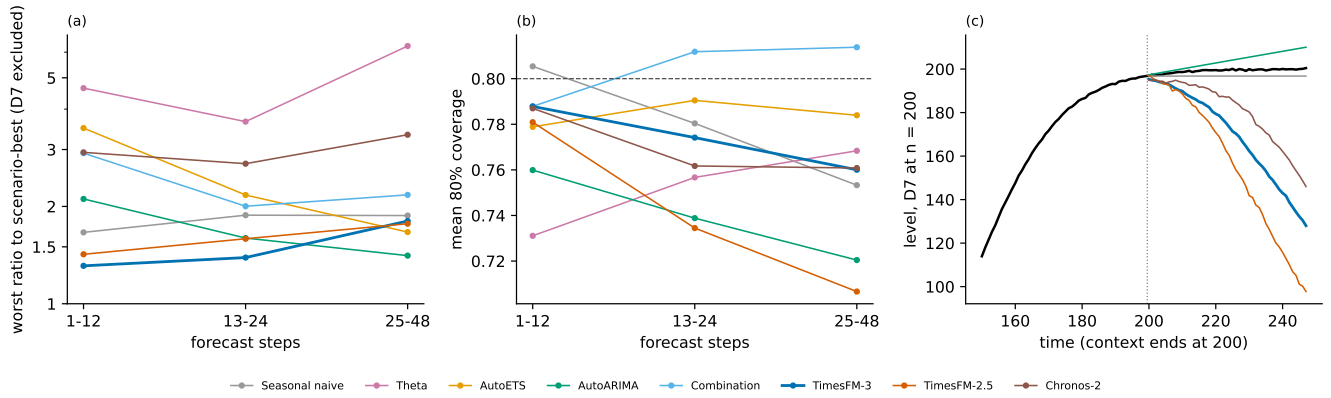


FIGURE S3 | Longer horizon ($H = 48$). (a) Worst ratio to the scenario-best method by block of forecast steps, D7 excluded. (b) Mean coverage of the nominal 80% intervals by block; the dashed line is the nominal level. (c) D7 at $n = 200$: mean of the 200 series (black) and mean point forecasts over the 48 steps.

TABLE S16 | Data observed after the documented training corpora: 101 FRED-MD series, January 2025 to June 2026.

Method	Mean MASE	Median MASE	Mean MASE, 2025-11 on	Coverage 80%	SPL
AutoARIMA	0.540	0.411	0.722	0.913	0.241
TiRex	0.556	0.439	0.746	0.864	0.229
Combination	0.556	0.445	0.756	0.880	0.250
AutoETS	0.572	0.455	0.774	0.908	0.262
Chronos-Bolt	0.582	0.505	0.776	0.887	0.238
Chronos-2	0.593	0.424	0.808	0.798	0.245
Theta	0.602	0.479	0.833	0.843	0.270
TimesFM-3	0.607	0.426	0.857	0.838	0.242
TimesFM-2.5	0.608	0.404	0.833	0.850	0.254
Seasonal naive	0.896	0.664	1.145	0.911	0.381

Note: FRED-MD, August 2026 vintage, raw levels; context January 2005 to December 2024, horizon 18. MASE scaled by the in-sample seasonal-naive error ($m = 12$). The later window (November 2025 to June 2026) post-dates the release of every model evaluated.

TABLE S17 | Scenarios in which TimesFM-3 is separated from each opponent: simultaneous against per-scenario intervals.

Opponent	Simultaneous (max- t)		Per scenario	
	TimesFM-3 better	Opponent better	TimesFM-3 better	Opponent better
AutoARIMA	8	4	11	9
Seasonal naive	17	7	22	8
Theta	23	0	28	1
AutoETS	17	0	25	3
Combination	15	2	27	3

Note: Median paired $\log_2$ ratio of MASE over $h = 1, \dots, 12$ in each of the 36 scenarios; 95% intervals from 2000 bootstrap resamples of the replications, simultaneous over the 36 scenarios (studentised maximum) or per scenario (percentile).